

Conditional Distribution :

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The conditional distributions are exactly analogous to the conditional probabilities of the type $P(A|B)$ or $P(B|A)$, where A and B are two events in a sample space.

Using the definition of conditional probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)} ; P(A) > 0$$

Replacing the events A and B by the random variables X and Y , respectively, we can define the conditional probability of Y for given X as follows

$$P(Y=y | X=x) = \frac{P(X=x) P(Y=y)}{P(X=x)}$$
$$= \frac{f(x, y)}{g(x)}$$

Where X and Y are discrete random variables.

The function $f(x, y) / g(x)$ is a function of y with x held fixed. We call this function conditional probability distribution of the discrete random variable Y given $X=x$. It satisfies all the properties of a probability distribution. Writing this probability distribution as $f(y|x)$, we have

$$f(y|x) = \frac{f(x, y)}{\sum_y f(x, y)} = \frac{f(x, y)}{g(x)}$$

; for $g(x) > 0$

Similarly, the conditional probability distribution of the discrete random variable X , given $Y=y$ is defined as

$$f(x|y) = \frac{f(x, y)}{\sum_x f(x, y)} = \frac{f(x, y)}{h(y)}$$

; for $h(y) > 0$

If X and Y are continuous random variables, then their corresponding conditional probability distributions will similarly be defined as follows:

$$f(y|x) = \frac{f(x,y)}{\int_{-\infty}^{\infty} f(x,y) dy} = \frac{f(x,y)}{g(x)}; g(x) > 0$$

$$\& f(x|y) = \frac{f(x,y)}{\int_{-\infty}^{\infty} f(x,y) dx} = \frac{f(x,y)}{h(y)}; h(y) > 0$$

If someone wishes to find the probability that the random variable X falls between a and b , when it is known that variable $Y=y$, we evaluate,

$$P(a < X < b | Y=y) = \begin{cases} \sum_{x=a}^b f(x|y), & \text{if } X \text{ is discrete} \\ \int_a^b f(x|y) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Example: Given the following joint distributions of the discrete random variables X and Y .

	X		
Y	0	1	2
0	$3/8$	$9/128$	$3/128$
1	$6/28$	$6/28$	0
2	$1/28$	0	0

Find $f(x|1)$, $f(y|1)$ & $P(X=0|Y=1)$.

Soln: By definition;

$$f(x|1) = \frac{f(x,1)}{h(1)} \quad \& \quad f(y|1) = \frac{f(y,1)}{g(1)}$$

$$\text{Now, } h(1) = \sum_{x=0}^2 f(x,1) = f(0,1) + f(1,1) + f(2,1) \\ = 6/28 + 6/28 + 0 = 3/7$$

Hence, the conditional distribution of X given

$Y=1$ is

$$f(x|1) = \frac{f(x,1)}{h(1)} = \frac{7}{3} f(x,1) \text{ for } x=0,1,2$$

Therefore,

$$f(0|1) = \left(\frac{7}{3}\right) f(0,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(1|1) = \left(\frac{7}{3}\right) f(1,1) = \frac{7}{3} \times \frac{6}{28} = \frac{1}{2}$$

$$f(2|1) = \left(\frac{7}{3}\right) f(2,1) = \frac{7}{3} \times 0 = 0$$

• Hence the conditional distribution of X , given $Y=1$ in tabular form is

x	0	1	2
$f(x 1)$	$1/2$	$1/2$	0

• Again

$$g(1) = \sum_{y=0}^2 f(1,y) = f(1,0) + f(1,1) + f(1,2) \\ = \frac{9}{28} + \frac{6}{28} + 0 = \frac{15}{28}$$

Hence, $f(y|1) = \frac{f(1,y)}{g(1)} = \frac{28}{15} f(1,y);$
for $y=0,1,2$

Therefore,

$$f(0|1) = \binom{28}{15} f(1,0) = \frac{28}{15} * \frac{9}{28} = \frac{3}{5}$$

$$f(1|1) = \binom{28}{15} f(1,1) = \frac{28}{15} * \frac{6}{28} = \frac{2}{5}$$

$$f(2|1) = \binom{28}{15} f(1,2) = \frac{28}{15} * 0 = 0$$

Hence, the conditional distribution of Y given $X=1$ is

$Y:$	0	1	2
$f(Y 1):$	$\frac{3}{5}$	$\frac{2}{5}$	0

Also,

$$P(X=0|Y=1) = \frac{f(0,1)}{h(1)} = f(0|1) = \frac{1}{2}$$

$$\begin{aligned} \Rightarrow P(X=0|Y=1) &= \frac{6/28}{3/7} \\ &= \frac{2/8}{4/8} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{or, } P(X=0|Y=1) &= f(0|1) \\ &= \frac{1}{2} \end{aligned}$$

Example : Consider the following probability distribution of X and Y .

$$f(x, y) = \frac{x+y}{21}; \quad x=1, 2, 3 \text{ \& } y=1, 2$$

(i) Obtain the marginal and conditional distributions of X and Y .

(ii) find $f(x|1)$ and

(iii) $P(X=2|Y=1)$.

Solⁿ : The marginal distributions of X and Y are

$$(i) \quad g(x) = \sum_{y=1}^2 f(x, y) = \sum_{y=1}^2 \frac{x+y}{21}$$

$$\Rightarrow g(x) = \frac{x+1}{21} + \frac{x+2}{21} = \frac{2x+3}{21}; \text{ for } x=1, 2, 3$$

$$\& \quad h(y) = \sum_{x=1}^3 f(x, y) = \sum_{x=1}^3 \frac{x+y}{21}$$

$$\Rightarrow h(y) = \frac{1+y}{21} + \frac{2+y}{21} + \frac{3+y}{21} = \frac{3y+6}{21}; \text{ for } y=1, 2$$

Thus the conditional distributions of X and Y are:

$$f(x|y) = \frac{f(x,y)}{h(y)} = \frac{(x+y)/4}{(3y+6)/4}$$

$$\Rightarrow f(x|y) = \frac{x+y}{3y+6} ; \text{ for } x=1,2,3 \text{ \& } y=1,2$$

And,

$$f(y|x) = \frac{f(x,y)}{g(x)} = \frac{(x+y)/4}{(2x+3)/4}$$

$$\therefore f(y|x) = \frac{x+y}{2x+3} ; \text{ for } x=1,2,3 \text{ \& } y=1,2$$

(ii) The conditional distribution of X for $Y=1$

is

$$f(x|1) = \frac{x+1}{3 \cdot 1 + 6} = \frac{x+1}{9} ; x=1,2,3$$

(iii) The conditional probability of X for given

Y is

$$P(X=2|Y=1) = \frac{2+1}{3 \cdot 1 + 6} = \frac{3}{9} = \frac{1}{3}$$

(Ans) $\rightarrow$

Example: Two random variables X and Y have the following joint probability density function:

$$f(x, y) = \begin{cases} \frac{1}{2} & ; 0 \leq x \leq y \leq 2 \\ 0 & ; \text{elsewhere} \end{cases}$$

(a) Find the marginal density of Y and hence the conditional density of X .

(b) Find also $P(X \leq 0.5 | Y = 1.5)$

Sol: (a) The marginal density of Y is given by

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$\Rightarrow h(y) = \int_0^y \frac{1}{2} dx = \frac{1}{2} [x]_0^y = \frac{1}{2} y$$

Thus,

$$h(y) = \begin{cases} \frac{1}{2} y & ; 0 \leq y \leq 2 \\ 0 & ; \text{elsewhere} \end{cases}$$

Thus for any $0 < y \leq 2$, the conditional density of X for given Y is

$$f(x|y) = \frac{f(x, y)}{h(y)}$$

$$= \frac{1/2}{1/2 y}$$

$$\Rightarrow f(x|y) = 1/y; \quad 0 \leq x \leq y$$

$$(b) P(X \leq 0.5 | Y = 1.5) = \int_{-\infty}^{0.5} f(x|y=1.5) dx$$

$$= \int_0^{0.5} \frac{1}{1.5} dx$$

$$= \frac{1}{1.5} [x]_0^{0.5}$$

$$= \frac{1}{3}$$

Ans

Note :

$$P(a < X < b | Y=y) = \int_a^b f(x|y) dx$$

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Example : Find the conditional density of Y given X for the following distribution

$$f(x, y) = \frac{6-x-y}{8} ; \quad 0 < x < 2, \quad 2 < y < 4$$
$$= 0 ; \text{ otherwise .}$$

Also compute $P(2 < Y < 3 | X=2)$

Solⁿ : The conditional density of Y given X is

$$f(y|x) = \frac{f(x, y)}{g(x)} , \text{ where}$$

$$g(x) = \int_{y=2}^4 f(x, y) dy = \frac{1}{8} \int_2^4 (6-x-y) dy$$

$$\Rightarrow g(x) = \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4$$

$$\Rightarrow g(x) = \frac{1}{8} \left[(24 - 4x - 8) - (12 - 2x - 2) \right]$$

$$= \frac{1}{8} [16 - 4x - 10 + 2x]$$

$$\therefore g(x) = \frac{1}{4} (3-x) ; \quad 0 < x < 2$$

Thus,

$$f(y|x) = \frac{(6-x-y)/8}{(3-x)/4}$$

$$\Rightarrow f(y|x) = \frac{6-x-y}{2(3-x)}, \quad 0 < x < 2, \quad 2 < y < 4$$

Now, when $x=2$,

$$P(2 < Y < 3 | X=2) = \int_{y=2}^3 f(y|2) dy$$

$$= \int_{y=2}^3 \frac{6-2-y}{2(3-2)} dy$$

$$= \frac{1}{2} \int_{y=2}^3 (4-y) dy$$

$$= \frac{1}{2} \left[4y - \frac{y^2}{2} \right]_2^3$$

$$= \frac{1}{2} \left[\left(12 - \frac{9}{2} \right) - \left(8 - 2 \right) \right]$$

$$= \frac{1}{2} \left(6 - \frac{9}{2} \right)$$

$$= \frac{3}{4} ; \quad 2 < y < 4$$

~~Ans~~

Independence of random variables:

Two random variables X and Y with marginal densities $g(x)$ and $h(y)$, respectively are said to be independent if and only if

$$f(x|y) = g(x)$$

$$\text{or } f(y|x) = h(y)$$

where $f(x|y)$ is the conditional density of X for a given Y and $f(y|x)$ is the conditional density of Y for a given X .

If X and Y are independent, then for any real numbers x and y it must be true that

$$f(x, y) = g(x) * h(y)$$

for all values of x & y .

Note:

$$\left\{ \begin{array}{l} P(A|B) = P(A) \text{ or } P(B|A) = P(B) \rightarrow \text{Method-1} \\ P(A \cap B) = P(A) * P(B) \rightarrow \text{Method-2} \end{array} \right.$$

Example:

Suppose X and Y have the following joint probability function

X			
Y	2	4	Row sum
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
Column Sum	0.40	0.60	1.00

Check whether X and Y are independent or not;

Solⁿ: (Method-1):

To check the independence of X and Y , we must compute $g(x)$ and $h(y)$ for all values of the random variables X and Y and equate to $f(x, y)$. These computations are shown below:

$$(i) f(2, 1) = 0.10; g(2) = 0.40; h(1) = 0.25;$$

$$\text{Hence, } g(2) * h(1) = 0.10 = f(2, 1)$$

$$(ii) f(4,1) = 0.15; g(4) = 0.60; h(1) = 0.25;$$

$$\text{Hence, } g(4) * h(1) = 0.15 = f(4,1)$$

$$(iii) f(2,3) = 0.20; g(2) = 0.40; h(3) = 0.50;$$

$$\text{Hence, } g(2) * h(3) = 0.20 = f(2,3)$$

$$(iv) f(4,3) = 0.30; g(4) = 0.60; h(3) = 0.50;$$

$$\text{Hence, } g(4) * h(3) = 0.30 = f(4,3)$$

$$(v) f(2,5) = 0.10; g(2) = 0.40; h(5) = 0.25$$

$$\text{Hence, } g(2) * h(5) = 0.10 = f(2,5)$$

$$(vi) f(4,5) = 0.15; g(4) = 0.60; h(5) = 0.25$$

$$\text{Hence, } g(4) * h(5) = 0.15 = f(4,5)$$

The above computations show that for all pairs of the values (x, y) of the random variables X and Y , $f(x, y) = g(x) * h(y)$.

Hence, the random variables are independent.

(Method-2):

An alternative way to check the independence of X and Y is done through computing the conditional probabilities of the type

$P(Y=i | X=j)$ for all i & j values.

Now,

$$(i) P(Y=1 | X=2) = f(1|2) = \frac{f(2,1)}{g(2)}$$

$$\cancel{P(Y=1)} = \frac{0.10}{0.40} = 0.25 = h(1)$$

$$(ii) P(Y=3 | X=2) = f(3|2) = \frac{f(2,3)}{g(2)} = \frac{0.20}{0.40} = 0.50 = h(3)$$

$$(iii) P(Y=5 | X=2) = f(5|2) = \frac{f(2,5)}{g(2)} = \frac{0.10}{0.40} = 0.25 = h(5)$$

$$(iv) P(Y=1 | X=4) = f(1|4) = \frac{f(4,1)}{g(4)} = \frac{0.15}{0.60} = 0.25 = h(1)$$

$$(v) P(Y=3 | X=4) = f(3|4) = \frac{f(4,3)}{g(4)} = \frac{0.20}{0.60} = 0.50 = h(3)$$

$$(vi) P(Y=5 | X=4) = f(5|4) = \frac{f(4,5)}{g(4)} = \frac{0.15}{0.60} = 0.25 = h(5)$$

So, from the above computations we can say X and Y are independent random variables.

Example :

For the following probability distribution, show that X and Y are not independent.

$$f(x, y) = \frac{x+y}{8} ; \quad 0 < x < 2, \quad 0 < y < 2$$

Solⁿ : (Method - 1):

$$g(x) = \int_{y=0}^2 f(x, y) dy ; \quad \text{for } 0 < x < 2$$

$$= \frac{1}{8} \int_{y=0}^2 (x+y) dy$$

$$= \frac{1}{8} \left[xy + \frac{y^2}{2} \right]_0^2$$

$$= \frac{1}{8} (2x + 2)$$

$$\therefore g(x) = \frac{1}{4} (x+1) ; \quad \text{for } 0 < x < 2$$

Also,

$$h(y) = \int_{x=0}^2 f(x, y) dx ; \quad \text{for } 0 < y < 2$$

$$= \frac{1}{8} \int_{x=0}^2 (x+y) dx$$

$$= \frac{1}{8} \left[\frac{x^2}{2} + xy \right]_0^2$$

$$= \frac{1}{8} (2+y)$$

$$\therefore h(y) = \frac{1}{4} (y+1) ; \text{ for } 0 < x < 2$$

$$\text{Thus, } g(x) * h(y) = \frac{1}{4} (x+1) \cdot \frac{1}{4} (y+1)$$

$$\Rightarrow g(x) * h(y) = \frac{1}{16} (x+1) (y+1) \neq f(x, y)$$

$$\therefore g(x) * h(y) \neq f(x, y)$$

Hence, x and y are not independent.

(Method-2)

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{(x+y)/8}{(x+1)/4}$$

$$\Rightarrow f(y|x) = \frac{1}{2} \left(\frac{x+y}{x+1} \right) \neq h(y)$$

$$\& f(x|y) = \frac{f(x, y)}{h(y)} = \frac{(x+y)/8}{(y+1)/4}$$

$$\Rightarrow f(x|y) = \frac{1}{2} \left(\frac{x+y}{y+1} \right) \neq g(x)$$

Since the conditional distributions are not equal to the marginal distributions, so the variables are not independent.

Example : Given the following joint density function of X and Y as follows:

$$f(x, y) = \begin{cases} 6xy^2, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Verify if X and Y are independent.

Solⁿ : The marginal density of X is

$$g(x) = \int_{y=-\infty}^{\infty} f(x, y) dy = 6 \int_0^1 xy^2 dy; \quad 0 \leq x \leq 1$$

$$\Rightarrow g(x) = 6 \left[x \frac{y^3}{3} \right]_0^1 = 2x; \quad 0 \leq x \leq 1$$

Similarly, the marginal density of Y is

$$h(y) = \int_{x=-\infty}^{\infty} f(x, y) dx = 6 \int_0^1 xy^2 dx; \quad 0 \leq y \leq 1$$

$$\Rightarrow h(y) = 6 \left[\frac{x^2}{2} y^2 \right]_0^1 = 3y^2; \quad 0 \leq y \leq 1$$

Hence,

$$g(x) * h(y) = 2x * 3y^2 = 6xy^2 = f(x, y)$$

$\therefore f(x, y) = g(x) * h(y)$ for all real numbers

Therefore, X and Y are independent.

Example: Given the following joint density function of X and Y as

$$f(x, y) = \begin{cases} 2; & 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

Show that X and Y are not independent.

Solⁿ: The marginal density function of X is given

by $g(x) = \int_{y=0}^1 f(x, y) dy$, for $0 \leq x \leq 1$

$$= \int_0^1 2 dy = 2y \Big|_0^1 = 2(1) - 2(0) = 2$$

$\therefore g(x) = 2$ ~~$g(x) = 2x$~~ $\therefore g(x) = 2$

The marginal density function of Y is given

by $h(y) = \int_{x=0}^1 f(x, y) dx = \int_0^1 2 dx = 2$

$$\therefore h(y) = 2$$

Hence,

$$f(x, y) \neq g(x) \cdot h(y)$$

Therefore, X and Y are not independent.